

SHUVAM PAL

+91 85830 54679 ✉ ishuvam.pal@gmail.com  Shuvam Pal  unichronic

Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Optimized an **AI-assisted workflow pipeline** for company-name suggestions, structured filing descriptions, and operator-ready drafts with validation before handoff.
- Built a resilient automation backend with retries, session recovery, public-site API lookups, and state validation, reducing manual filing time by **85%**.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python medical-image segmentation tooling for **95 brain subparts**, handling preprocessing, orientation correction, model inference, generated masks, and large-volume MRI data paths.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Debugged model-output regressions across conformation, orientation, label mapping, PyTorch/ONNX-style inference, and binary mask generation in a mature open-source codebase.

Open Source

Contributions 

- Contributed patches across Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside maintainer review processes for reliability, security, and integration quality.

Projects

Seaweed: Contest Platform | *Go, TypeScript, PostgreSQL, Judge0, Firebase Auth, Redis, Docker/Kubernetes* **GitHub** 

- Built an **education-style coding assessment platform** for contest registration, problem access windows, code submissions, automated judging, live ranked leaderboards, and shortlist/admin workflows.
- Designed the **Judge0 evaluation pipeline** with async Go dispatch, per-test runtime/memory telemetry, PostgreSQL ranking updates, and materialized-view leaderboard refreshes for **500+ concurrent users** and **sub-5s P95 verdict latency**.

Swish Support System | *Python, FastAPI, LLM APIs, PostgreSQL, Redis, Langfuse, Pytest*

GitHub 

- Built an **AI feedback backend** where LLM assessments classify user intent, severity, ambiguity, requested resolution, and evidence usefulness while deterministic services control policy-sensitive actions.
- Added response-quality checks, copy-contract guardrails, LLM-judge scoring, Langfuse traces, Redis session context, and regression tests so generated feedback stayed useful, grounded, and reviewable.

Penny Lane Capital | *Python, Agno, Pydantic, SQLite, Embeddings, Vector Memory, JSONL*

GitHub 

- Built a **multi-agent research workflow** with analyst roles, debate, risk review, portfolio approval, typed Pydantic state, SQLite checkpoints, and inspectable JSONL traces.
- Implemented **vector memory and semantic lesson retrieval** with local embeddings, reward-scored loops, and validation harnesses so agent behavior and retrieved context could be evaluated across runs.

Technical Skills

Languages: Python, Go, TypeScript, C++

Frameworks: FastAPI, Echo, Node.js, PyTorch, ONNX Runtime, Agno, Pytest

Tools: PostgreSQL, SQLite, Redis, Firebase Auth, Judge0, Docker, Kubernetes, AWS, GCP, GitHub Actions, Langfuse, Prometheus, OpenTelemetry